

Dependent events and increasing probability

If the first card chosen from a pack of 40 is not one of the 4 pink cards, then the probability of the next card being pink grows to 4 out of the remaining 39 cards ($4/39$). In this example, the probability of a pink card being the next to be picked grows to a certainty with each non-pink card picked.

The first card has been chosen and is not pink. The next card to be picked will be chosen from the remaining 39 cards.



chances of next card being pink

$$\frac{4}{39}$$

cards left

◁ **The chances of the** next card being pink are 4 out of 39 ($4/39$). This is because none of the 4 pink cards were picked so there are 4 still left in the pack.

The first 12 cards have been chosen, none of which were pink. The next card to be picked will be chosen from the remaining 28 cards.



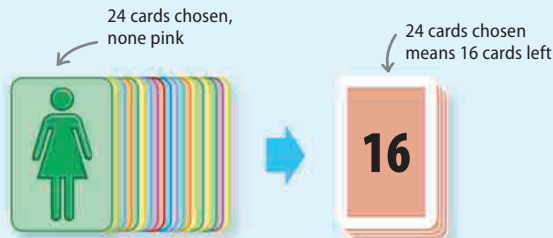
chances of next card being pink

$$\frac{4}{28}$$

cards left

◁ **The chances of the** next card being pink are 4 out of 28 ($4/28$). With none of the 4 pink cards picked there are still 4 left in the pack.

24 cards have been chosen and none of which were pink. The next card to be picked will be chosen from the remaining 16 cards.



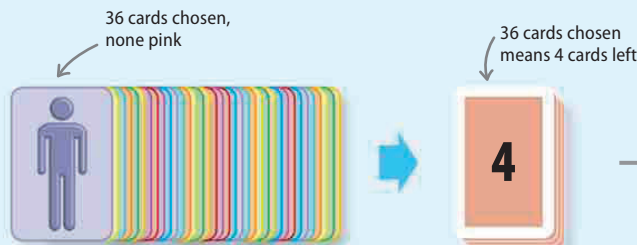
chances of next card being pink

$$\frac{4}{16}$$

cards left

◁ **The chances of the** next card being pink are 4 in 16. With none of the 4 pink cards picked there are 4 left in the pack.

The first 36 cards have been chosen. None of them were pink. The next card to be picked will be chosen from the remaining 4 cards.



chances of next card being pink

$$\frac{4}{4}$$

cards left

◁ **The chances of the** next card being pink are 4 in 4 ($4/4$), or a certainty. With none of the pinks chosen there are 4 left in the pack.